GAMING ADDICTION RISK PREDICTION

Using Gradient Boosting & XGBoost

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أكاديمية طويق
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72.5%
Best Model Accuracy

1,000
Dataset Records

4 Classes

Presentation Structure

Three-part delivery across the team

PART 1

Algorithm Explanation

Mathematical and statistical foundations of Gradient Boosting and XGBoost

PART 2

Data & Exploration

Dataset overview, EDA visualizations, preprocessing pipeline

PART 3

Evaluation Metrics

Model results, confusion matrices, comparison, and feature importance

01

Algorithm Explanation

Mathematical and statistical foundations of Gradient Boosting & XGBoost

What is Gradient Boosting?

Definition

Gradient Boosting is an ensemble learning method that combines many weak learners (typically shallow decision trees) sequentially. Each new tree is trained to correct the residual errors of the combined ensemble so far, using gradient descent on a loss function to guide improvement.

Unlike bagging (parallel trees), boosting builds trees one after another, each focusing on where the model currently fails.

Key Properties

- Sequential ensemble — trees are added one at a time
- Loss-driven — minimizes a differentiable loss function
- Additive model — final prediction is sum of all tree outputs
- Regularized — learning rate shrinks each tree's contribution

How Gradient Boosting Works — The Math

1

Initialize Model

$F_0(x) = \text{argmin}_n \Sigma L(y_i, \gamma)$

Start with a constant prediction that minimizes the loss over all training samples.

2

Compute Residuals

$r_{im} = -\partial L(y_i, F(x_i)) / \partial F(x_i)$

Calculate the negative gradient of the loss function — these pseudo-residuals tell each new tree where the current model is wrong.

3

Fit a New Tree

$h_m(x)$ fitted to $\{(x_i, r_{im})\}$

Train a shallow decision tree on the pseudo-residuals. The tree learns to predict the direction and magnitude of errors.

4

Update Model

$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x)$

Add the new tree's predictions scaled by learning rate η to the ensemble. Repeat M times.

XGBoost — Extreme Gradient Boosting

XGBoost extends Gradient Boosting with a regularized objective function and engineering optimizations that make it faster, more accurate, and more robust.

Regularized Objective Function

$$Obj = \Sigma L(y_i, \hat{y}_i) + \Sigma[\gamma T + \tfrac{1}{2}\lambda w^2]$$

Loss term drives accuracy | Regularization term ($\gamma T + \lambda w^2$) penalizes tree complexity to prevent overfitting

Second-Order Gradients

Uses both first and second derivatives (Hessian) for more precise split finding

Column Subsampling

Randomly selects features per tree and per level, reducing overfitting

Sparsity-Aware Splits

Natively handles missing values by learning optimal default split directions

Weighted Quantile Sketch

Approximate split finding for scalability on large datasets

When to Use | Advantages | Limitations

WHEN TO USE

- Multi-class classification tasks
- Structured/tabular datasets
- When high accuracy is priority
- Imbalanced class distributions
- Complex feature interactions exist

ADVANTAGES

- Handles non-linear relationships
- Built-in feature importance
- Robust to outliers and noise
- Supports class weighting
- Strong generalization ability

LIMITATIONS

- Prone to overfitting if untuned
- Slower training than simple models
- Sensitive to hyperparameters
- Less interpretable than trees
- Requires careful regularization

02

Dataset & Exploration

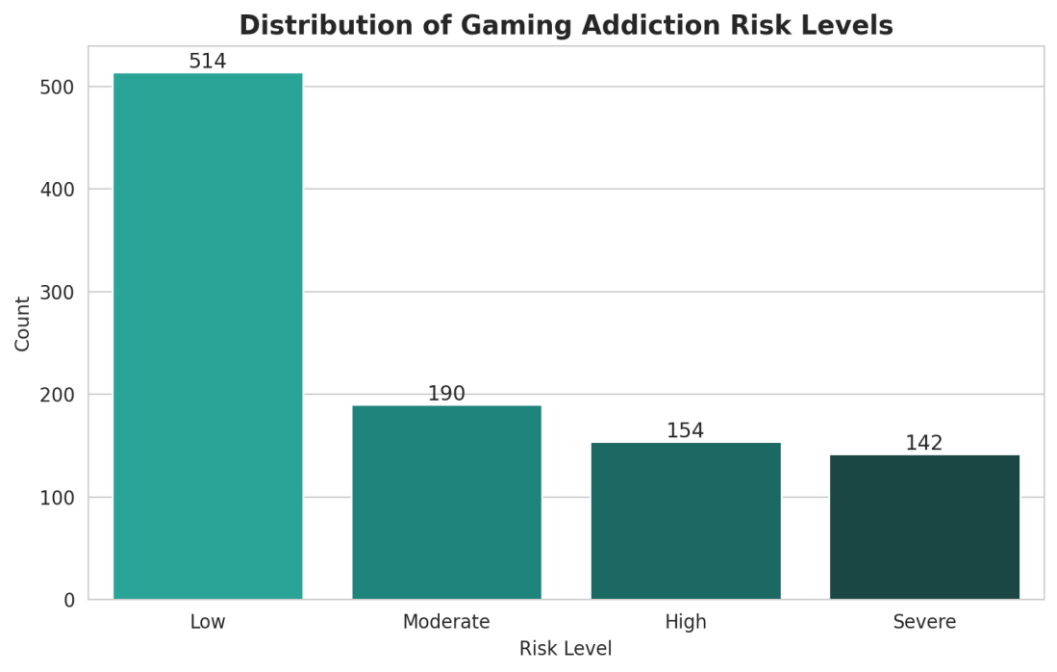
Understanding the Gaming and Mental Health dataset through EDA

Dataset Structure & Features

1,000	27	4 Classes	2
Records	Columns	Target	Missing Cols

Feature Category	Examples	Type
Demographics	age, gender	Numeric / Categorical
Gaming Behavior	daily_gaming_hours, game_genre, platform	Mixed
Sleep & Health	sleep_hours, sleep_quality, eye_strain	Numeric / Categorical
Social & Academic	social_isolation_score, grades_gpa	Numeric
Target Variable	**gaming_addiction_risk_level**	**Low / Moderate / High / Severe**

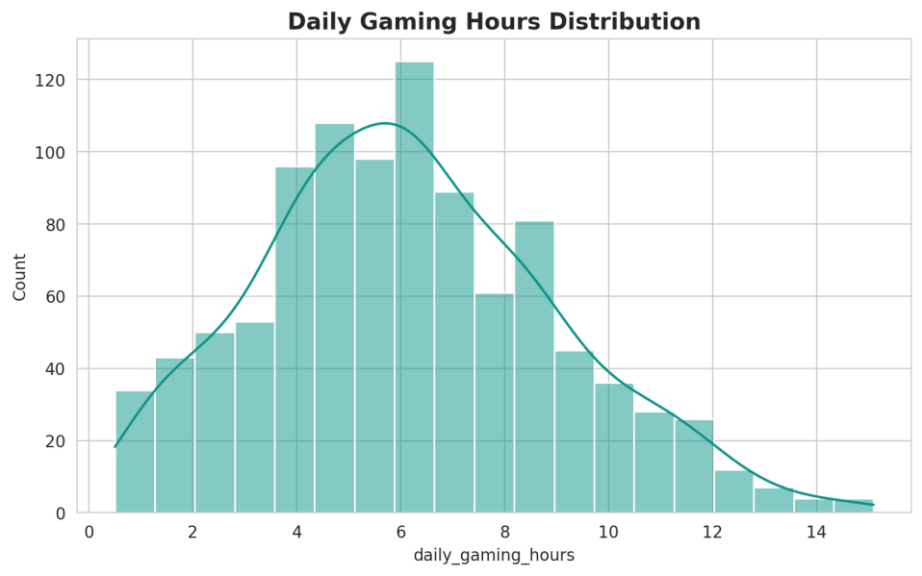
EDA — Target Variable Distribution

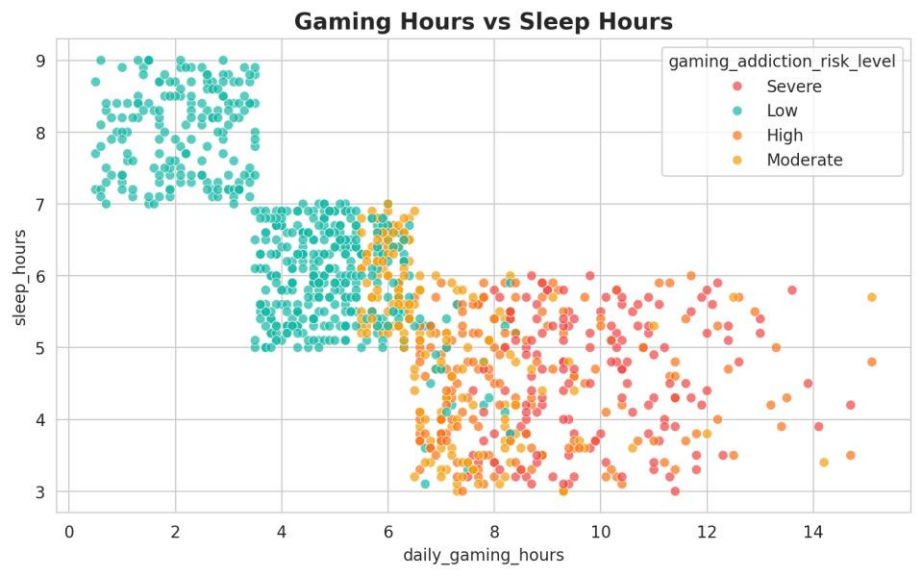


Key Insight

The dataset is clearly imbalanced — the Low risk class dominates, while Severe cases are under-represented. This imbalance can bias the model toward the majority class, requiring techniques like class weighting or SMOTE to ensure fair predictions across all risk levels.

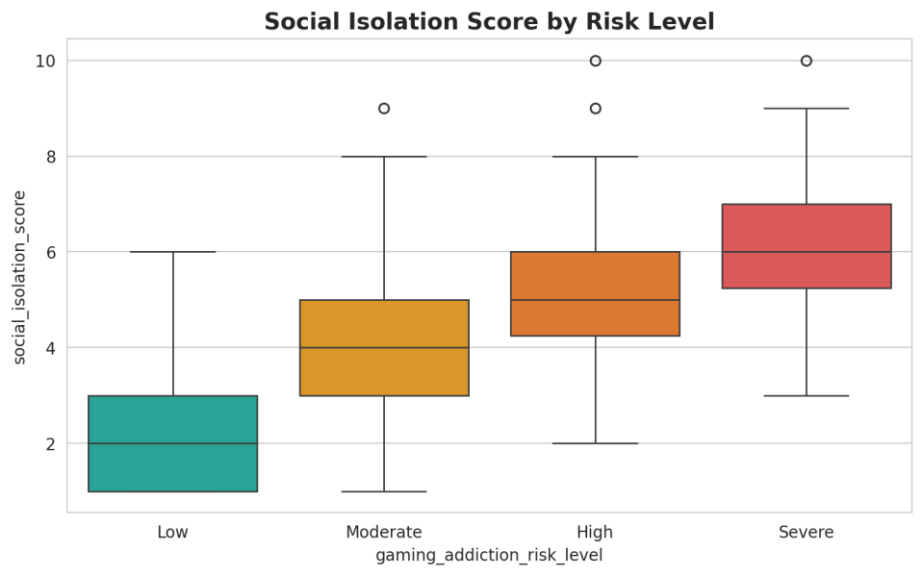
EDA — Gaming Hours & Sleep Patterns

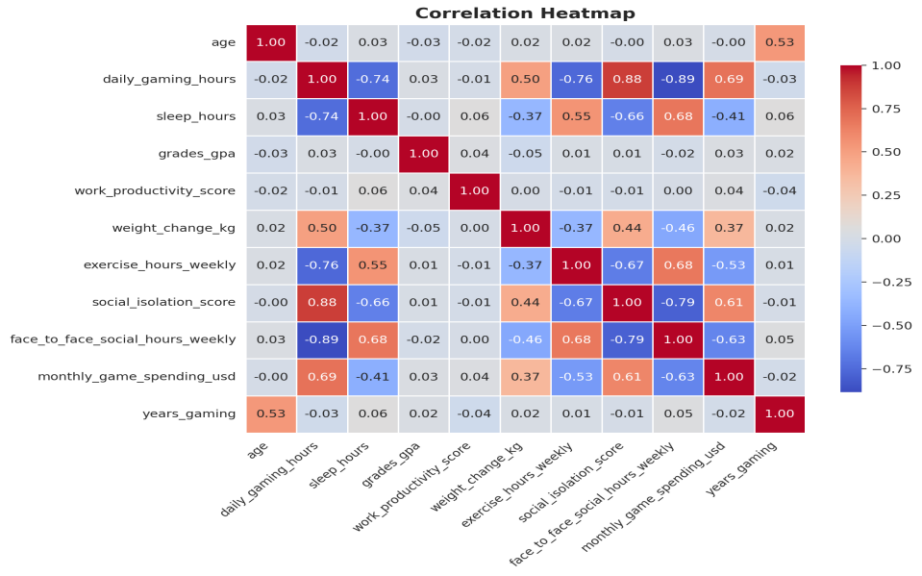




Most users game 4-8 hours/day (right-skewed). A clear negative relationship exists between gaming hours and sleep — higher gaming is associated with higher addiction risk, making both features strong predictors.

EDA — Social Isolation & Correlation Analysis





Social isolation scores increase with addiction severity. The heatmap reveals interconnected behavioral features — gaming addiction is linked to a combination of factors, not a single predictor.

Data Preprocessing Pipeline

1

Handle Missing Values

grades_gpa and work_productivity_score filled with median — robust to outliers, preserves distribution.

2

Remove Data Leakage

Dropped withdrawal_symptoms, loss_of_other_interests, continued_despite_problems — these define addiction, not predict it.

3

Remove Irrelevant Columns

Dropped record_id (identifier) and primary_game (high-cardinality, adds noise).

4

Encode Categoricals

One-Hot Encoding applied to gender, game_genre, gaming_platform, mood_state, etc. via ColumnTransformer.

5

Train/Test Split

80/20 stratified split (random_state=42). Stratification ensures class proportions are preserved.

6

Label Encode Target

Target encoded: High=0, Low=1, Moderate=2, Severe=3 for multi-class classification.

03

Model Evaluation

Performance metrics, confusion matrices, comparison, and feature importance

Understanding Evaluation Metrics

Accuracy

$$(TP + TN) / Total$$

Overall proportion of correct predictions across all classes. Can be misleading for imbalanced datasets.

Precision

$$TP / (TP + FP)$$

Of all samples predicted as class X, how many actually belong to class X? Measures false positive rate.

Recall

$$TP / (TP + FN)$$

Of all actual class X samples, how many did the model correctly identify? Measures false negative rate.

F1 Score

$$2 \times (P \times R) / (P + R)$$

Harmonic mean of precision and recall. Balances both metrics — especially important for imbalanced classes.

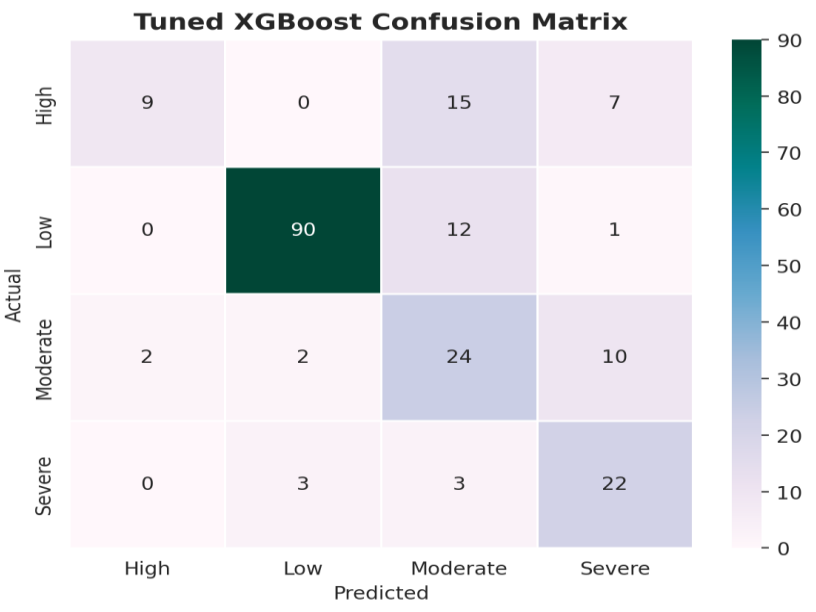
Baseline Model Results

Model	Accuracy	Precision	Recall	F1 Score
Gradient Boosting	69.0%	70.6%	69.0%	68.8%
XGBoost	67.5%	67.4%	67.5%	67.2%
Tuned XGBoost	72.5%	73.7%	72.5%	72.5%
KNN (k=5)	66.0%	62.8%	66.0%	63.8%

Key Findings

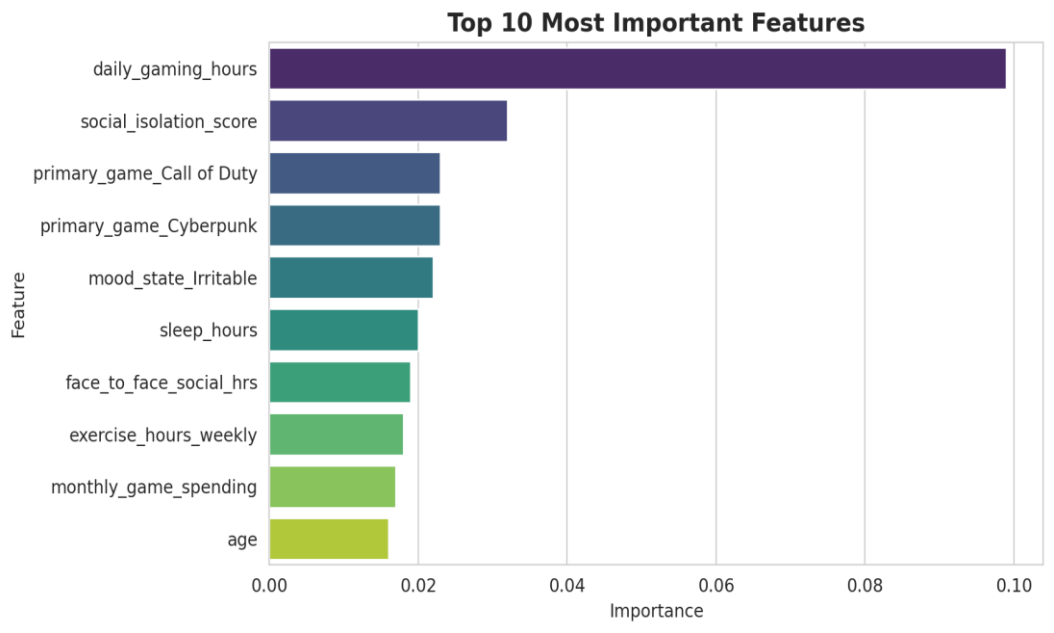
Tuned XGBoost achieved the best performance across all metrics (72.5% accuracy, 72.5% F1). Hyperparameter tuning with RandomizedSearchCV and class weighting improved XGBoost by +5% accuracy over the baseline. Tree-based models consistently outperformed KNN, indicating that distance-based methods are less suitable for this structured dataset with high-cardinality encoded features.

Model Comparison & Confusion Matrix



The confusion matrix shows Tuned XGBoost predicts Low risk reliably (90/103 correct) but struggles more with High and Moderate classes due to class imbalance and feature overlap between adjacent risk levels.

Feature Importance — What Drives Addiction Risk?



Top Predictors

daily_gaming_hours 9.9%

Dominant predictor — gaming behavior matters most

social_isolation_score 3.2%

Social withdrawal strongly linked to higher risk

sleep_hours 2.0%

Less sleep correlates with increased addiction

exercise_hours_weekly 1.8%

Physical activity serves as a protective factor

Behavioral features (gaming hours, social isolation) are far more impactful than demographic ones (age, gender), confirming that addiction risk is driven by modifiable habits rather than fixed traits.

Key Insights & Conclusion

1 Behavioral Factors Dominate

Daily gaming hours is the single most important predictor, followed by social isolation score. Demographic variables contribute relatively little.

2 Tuning Matters Significantly

Hyperparameter optimization with RandomizedSearchCV improved XGBoost accuracy from 67.5% to 72.5% — a meaningful gain for a real-world classification task.

3 Class Imbalance Is a Challenge

The Low risk class dominates the dataset. Using sample weights partially mitigated this, but minority classes (High, Severe) remain harder to predict.

4 Tree Models Outperform KNN

Gradient Boosting and XGBoost consistently beat KNN, confirming tree-based ensembles handle structured tabular data with encoded categoricals more effectively.

Thank You

Any Questions?

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72.5%
Best Accuracy

72.5%
Best F1 Score

1,000
Records

4
Models Compared